How PeX Resolves a Payment Exception

From exception event to governed resolution and final business outcome.

1. RESOLUTION JOURNEY · *Detection to Reasoning*

- **1. Ingest & Normalize:** Ingests raw MT/MX/ISO break from Payment Hub; normalizes into canonical PaymentExceptionCase.
- **2. Classification:** Maps break to 1 of 41 canonical profiles; derives root cause, missing fields, and failure codes.
- **3. Evidence Retrieval:** Asynchronously queries external systems (DDA status, client directory, SWIFTRef) via read-only ICDs.
- **4. AI Hypothesis:** Model Gateway evaluates candidate repairs and outputs structured ReasoningResult with confidence scores.

2. CONTROLLED DECISION · *Authority & Boundary*

- **Deterministic Rules:** 100% deterministic rule engine validates proposed repair against clearing schemes. Rules ALWAYS override AI.

CORE GOVERNANCE PRINCIPLE

"AI recommends. Controls validate.

Humans authorize. Systems execute."

- **Sanctions Re-Check:** Automated rescreening against watchlists to verify no new compliance risk is introduced.
- **Human-in-the-Loop:** Operator reviews side-by-side evidence, deterministic checks, and AI rationale on an intuitive canvas.
- **Maker-Checker SoD:** Hard $0 threshold (Maker ≠ Checker). Dual-Authorization required for transfers ≥$250k.

3. EXECUTION & OUTCOME · *Settlement & Closure*

- **Contract Generation:** Generates schema-compliant repair message (e.g., ISO 20022 pacs.008 amendment, camt.029 resolution).
- **Clearing Dispatch:** Securely dispatches packet to clearing rail via bank-controlled egress gateway; logs transport ACK (DISPATCHED).
- **Final Settlement:** Case transitions to SETTLED ONLY upon authoritative receipt of pacs.002 ACCP or internal DDA posting.
- **Merkle Audit Sealing:** Cryptographically seals complete 20-step lifecycle into immutable SHA-256 audit log.

GOVERNED LIFECYCLE ► EVENT ► INTELLIGENCE ► DETERMINISTIC CONTROL ► HUMAN AUTHORIZATION ► CONTROLLED EXECUTION ► FINAL SETTLEMENT ► MERKLE AUDIT

CORE TAKEAWAY: PeX closes the loop — from exception detection through investigation, governed decision, controlled execution, and final settlement.